Roll No.: DL.AI.UH.A12MIHD

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B.Tech. AI/DS Mid-Term Examinations – October 2025

First Semester

Artificial Intelligence & Data Science

Principles of Quantum Information Processing

Duration: Two hours | Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Explain the fundamental concepts of quantum information such as qubits, superposition, entanglement, and measurement, and visualize them using computational tools.
CO02	Design and analyze simple quantum circuits using single- and multi-qubit gates, and implement basic algorithms like Deutsch–Jozsa and Grover's search.
CO03	Apply quantum machine learning techniques and hybrid models to illustrative problems in medicine, including imaging, genomics, and drug discovery.
CO04	Evaluate the potential, limitations, and ethical implications of quantum technologies in healthcare and artificial intelligence.

Answer all questions

Part A

1) Define a Hilbert space and state two properties that make it suitable for describing quantum states.

[2] [CO01][BTL 1]

2) State the Born rule in quantum mechanics and explain its significance in terms of measurement outcomes of a quantum system.

[2] [CO01][BTL 2]

3) Are the gate combinations $H + S$ and $H.S$ valid quantum evolution operators or observables. where H is the Hadamard gate and $S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$?

[2] [CO02][BTL 4]

4) Define a Bloch sphere with diagram and situate the placement of state $|\psi\rangle = \frac{1}{2}|0\rangle + i\frac{\sqrt{3}}{2}|1\rangle$ on it.

[2] [CO01][BTL 2]

5) For a CNOT gate with qubit 1 as control and qubit 2 as target, compute the transformation of the operators X_1 and X_2 under conjugation by the gate: find CX_iC for $i = 1, 2$.

[2] [CO02][BTL 4]

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Part B

6) Discuss quantum entanglement using Bell states $|\psi_{\pm}\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle)$ and $|\phi_{\pm}\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$. Discuss how the entanglement in Bell states changes if we selectively apply X, Y or Z gate to the first or the second qubit.

[5] [CO02][BTL 5]

7) A quantum state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{\frac{i\pi}{4}}|1\rangle)$ is given. Design a measurement basis that can reveal the relative phase of this qubit. Explain why your chosen basis is suitable. State the expected measurement outcomes and their probabilities.

[5] [CO01][BTL 6]

8) State the no-cloning theorem, demonstrating mathematical understanding.

[5] [CO01][BTL 2]

9) Define teleportation with a diagram and explain the steps involved in teleporting an arbitrary single qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ from Alice to Bob.

[5] [CO02][BTL 3]

Part C

10) Let us define a new gate

$$U = \frac{1}{2}\begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}$$

Is this gate an evolution operator or an observable? Why?

[5] [CO01][BTL 3]

If we start with the state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, and pass it through two such gates U one after the other, what would be the output state?

[2] [CO01][BTL 5]

Determine a two-qubit quantum circuit that will, on input $|00\rangle$, produce output

$$|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

[3] [CO01][BTL 6]

11) Describe the Deutsch algorithm. With the help of a quantum circuit diagram, explain the sequence of operations and the measurement outcome.

[8] [CO02][BTL 4]

Briefly explain how quantum algorithms exhibit parallelism using the principle of superposition.

[2] [CO02][BTL 4]

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